EcGFp5 and EcMasFp5: desk evaluation of the four atlas leads

Date. 2026-10-31. **Question (owner).** "Evaluate the EcGFp5 leads": the four atlas rows `LEAD-ICP-gfp5-symmetrized-gaudry`, `LEAD-ICP-gfp5-odd-char-cover`, `LEAD-ICP-gfp5-oa-sdhp-jv` and `LEAD-ICP-gfp5-rho-reference-audit` (crypto-autoresearcher `knowledge/leads/ecdlp/ic-prime/`), under goal `GOAL-GFPN-380702`. **Status.** Desk evaluation. **Nothing here is measured, nothing is an improvement and nothing is a security claim about a deployed curve.** Every cost below is a model built from cited constants. The container's disk allowance was nearly spent, so nothing was built or run beyond small analysis scripts and one standalone Rust check.

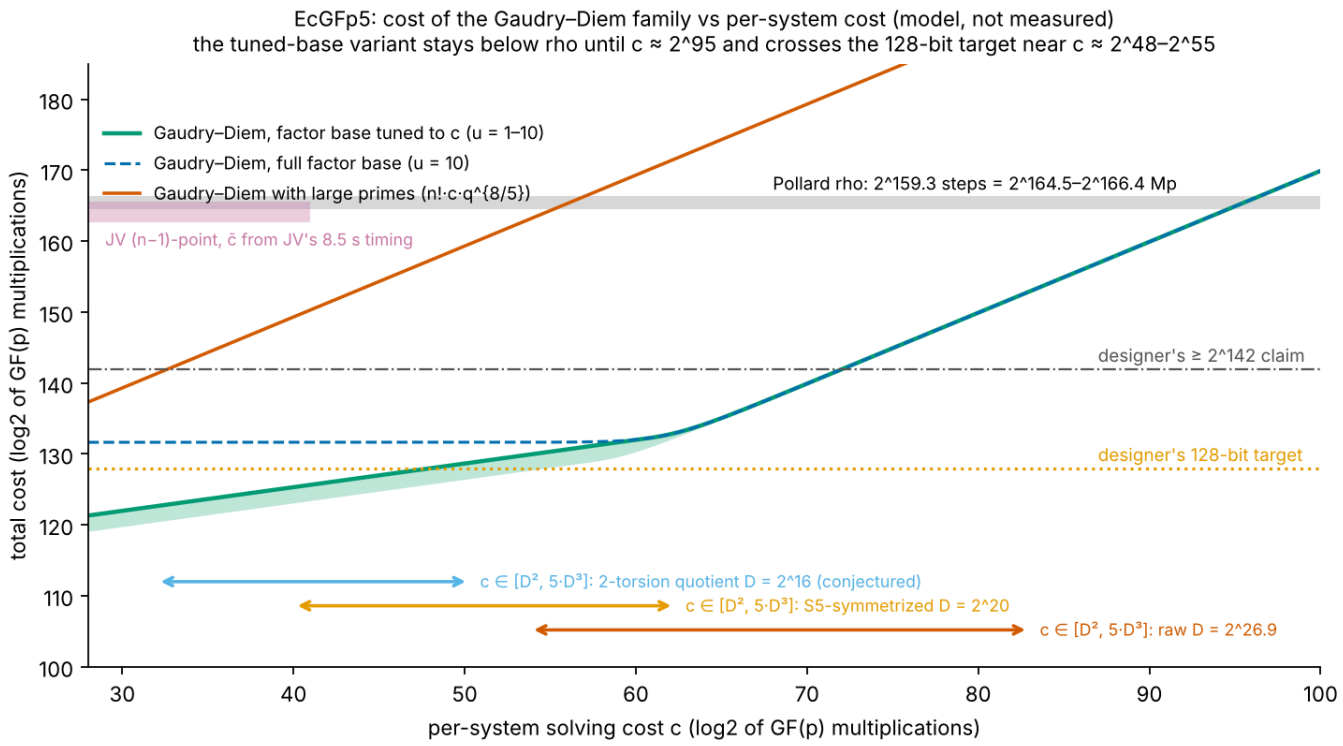
1. Answer in seven lines

1. **The design note's "≥ 2^142" is not a lower bound for the Gaudry–Diem family.** It multiplies the large-prime system count p^{2−2/5} = 2^102.4 by a per-system cost c, without the 5! = 120 inverse decomposition probability. Large primes are the *worst* rebalancing when each system is expensive.
2. **The cheapest member is the basic algorithm with a factor base tuned to c.** Two independent derivations (mine, and evaluator G1's) give a total that scales as **$c^{\{1/3\}: \text{about } 2\{115.5 + \log_2(c)/3\}}$** 64-bit word operations (G1). For the repository's own per-system bands (c between D^2 and $5 \cdot D^3$) that is **≈ $2^{126-2}136$** for D ≤ 2^20, i.e. at the designer's 128-bit target, not 14 bits above it (§3).
3. **In raw time every plausible variant is 32–45 bits below rho** (rho ≈ 2^159.3 steps ≈ 2^169 word operations). It stays below rho until c ≈ $2^{94-2}97$, beyond any per-system band.
4. **Memory is the barrier, not time.** The tuned factor base has $2^{57-2}63$ columns; linear algebra and relation storage need about **$2^{64-2}71$ bytes** against $2^{35-2}45$ for rho with distinguished points. Capped at a petabyte, the method is about 2^27 *above* rho (G1). So rho remains the better attack in any memory-aware cost model.
5. **Symmetrization matters less than the lead assumed.** Because the total scales as c^{1/3}, cutting the ideal degree from 2^20 to 2^16 buys only about 3–4 bits.
6. **The decisive unknown is still c(5, p)**, the cost of one complete five-variable solve, which the repository's C1 ladder never measured (it died at an 11 GiB cap; Joux–Vitse report the same memory wall). A frozen experiment with a 512 GiB envelope is specified in §5.
7. Verdicts: symmetrized Gaudry **REVISE** Joux–Vitse (n−1) for the full discrete log **CLOSE** (within ±2.5 bits of rho at ≈ 2^71 bytes); odd-characteristic cover **CLOSE** (genus 65 and 97, ≥ 2^206); oracle-assisted SDH **REVISE** (2^63 queries, no exposed oracle found); rho reference stays closed.

2. What was already known (read, not re-derived)

- **Primary source.** Pornin, ePrint 2022/274, frozen in crypto-autoresearcher `inputs/PORNIN-2022-274-ECGFP5/paper_fulltext.md`, lines 113–152: "Gaudry's attack entails computing about p^{2−2/5} ≈ 2^{102.4} systems … D … close to 2^{k(k−1)} = 2^{20} … unlikely to go below O(D^2) ≈ 2^{40} … a total theoretical complexity of at least 2^{142} … Joux and Vitse's variant has cost O(C p^2) … again above the 128-bit level." The target is 128-bit security, not rho's ≈ 2^160.
- **Joux–Vitse**, ePrint 2010/157, frozen in `inputs/JOUX-VITSE-2010-157/paper_fulltext.md`: Gaudry–Diem relation search n!·c(n,q)·q/2 and linear algebra Õ(n q^2) (lines 352–361); large primes give n!·c(n,q)·q^{2−2/n} (lines 362–367); the (n−1)-point variant costs (n−1)! ·c̃(n,q)·q^2/2 (lines 418–436); a 4-point decomposition over a 32-bit field takes 8.5 s on a 2.6 GHz Core 2 (lines 589–591); and "we have not been able to successfully run one complete relation search with [Gaudry–Diem's] method, as the requested memory exceeded the capacity of our personal computer" (lines 585–587).
- **The repository's C1 ladder** (`experiments/EXP-GFPN-05ff43/analysis.md`, decision DEC-20260923-d86ff2, inconclusive): no five-summand ideal degree was measured; both arms died at an 11 GiB address-space cap in the degree-33 F4 round at p′ = 4111. Its review (K7) found that the design note's degree-16, D ≈ 2^20 system is already the S5-symmetrized one (raw mixed volume 5!·16^5 ≈ 2^26.9), and re-anchored the bands to $2^{134.4-2}162.6$ (model only).
- **Open problem** `KN-OPEN-9b4a2b`: a genuine $\mathbb{F}_p$-rational (Z/2)^4 ⋊ S5 quotient on the double-odd curve is conjectured to have D = 2^16; seen only in toy square analogues.

3. The cost landscape



Cost landscape

My model is cost_landscape.py (unit: one F_p multiplication, Mp; constants and line references in its docstring; output cost_landscape.md). Evaluator G1 derived the same structure independently in 64-bit word operations (cards/g1.md, scripts in cards/g1_scripts/). With a factor base of B abscissae out of the L = p/2 that carry a point, a decomposition needs n!·(L/B)^n trials and about B relations are needed, so

$f(B) = c \cdot 5! \cdot L^5 / B^4 + \kappa \cdot B^2$, optimum $B^* \propto c^{1/6}$, $f(B^*) \propto c^{1/3}\,\kappa^{2/3}$.

G1's table ($\kappa \approx 960$ word operations per matrix entry, Wiedemann with normal equations; c in F_p multiply-adds at 4 words each; rho = 2^169.0 words):

log2 c (band)	design-note style p^{1.6}·c	full base	B*	tuned base	single large prime	memory, bytes
32 (D = 2^16, D²)	2^136.4	2^135.9	2^57.8	**2^126.2**	2^124.8	2^65.9
40 (D = 2^20, D²)	2^144.4	2^135.9	2^59.2	**2^128.8**	2^127.7	2^67.3
50 (D = 2^16, 5D³)	2^154.4	2^135.9	2^60.8	**2^132.2**	2^131.4	2^69.0
62.3 (D = 2^20, 5D³)	2^166.7	2^136.3	2^62.9	**2^136.3**	2^137.3	2^71.0
83 (D = 2^26.9, 5D³)	2^187.4	2^154.9	L	**2^154.8**	2^158	2^71.1

My model, with a cheaper linear-algebra constant (20–200 word operations per entry, which assumes a subquadratic Berlekamp–Massey), puts the same rows about 1.5–4 bits lower; the two agree on everything that decides a verdict. The sensitivity is $T \propto \kappa^{2/3}\,u^{1/3}$: across the plausible unit range the 2^128 crossing moves between log2 c ≈ 34 and ≈ 55, and rho moves by about one bit. If the designer's "128-bit" means group operations (≈ 2^137.6 words), the crossing moves to log2 c ≈ 65 (G1).

F4 can dominate FGLM. G1's dense Hilbert-function bound gives c_F4 ≈ 2^64.5 for the S5 system, above its FGLM figure, and $2^{51-2}54$ for an assumed quotient shape. That puts the tuned total at about $2^{132.5-2}133.6$, above 128. The quotient's shape is assumed, not committed.

4. Verdicts

lead	verdict	reason	next step
`LEAD-ICP-gfp5-symmetrized-gaudry`	**REVISE**	Its frame ("lower the design note's ≥ 2^142 floor") compares with the wrong member of the family; the right comparator is the tuned base, at or near 2^128. A D-drop from 2^20 to 2^16 is worth only 3–4 bits.	Measure c, not D, on the quotient (§5).
`LEAD-ICP-gfp5-oa-sdhp-jv`, (n−1) full-ECDLP half	**CLOSE**	$(n-1)!\cdot\tilde{c}\cdot q^2/2 = 2^{131.6}\cdot\tilde{c}$ with $\tilde{c} \approx 2^{33.5-2}34.4$ cycles from two anchors lands within ±2.5 bits of rho, at ≈ 2^71 bytes; it loses to rho for any B ≤ 2^62.5. Obstruction: memory, not arithmetic.	none

`LEAD-ICP-gfp5-oa-sdhp-jv`, oracle-assisted SDH half	**REVISE**	About p/2 = 2^63 oracle queries and 2^68.3 bytes of stored answers, then about 24p = 2^68.6 four-point decomposition trials per target (JV §5; G2). A protocol-level condition, not an ECDLP weakness; no deployment found exposes a static-DH oracle (signatures only).	Record the classification; defer measuring the per-trial cost unless a persistent [d]X service appears.
`LEAD-ICP-gfp5-odd-char-cover`	**CLOSE**	Canonical (2,...,2) cover has F_2-rank m = 5 on both curves: genus 65 (EcGFp5) and 97 (EcMasFp5), non-hyperelliptic. The cheap genus-5 class needs full rational 2-torsion and 4 | #E, impossible for orders 2n and n across both isogeny classes. Best constructed cover genus 33; index calculus there is ≥ 2^206 against rho 2^159.3.	Lift the source-freeze block by recording this computation; the Iijima–Momose–Chao text remains unread.
`LEAD-ICP-gfp5-rho-reference-audit`	stays **closed_known**	Reproduced: 2^159.33 steps with negation. Priced here at 37–130 Mp per step from Pornin's benchmark table (1,328.5-cycle addition, 10.18-cycle F_p multiplication), $2^{164.5-2}$166.4 Mp; G1 prices 800 words per step, 2^169.0 words.	Add the unit table to the lead's evidence.

4.1 Covers and oracle-assisted SDH (evaluator G2)

Full card: cards/g2.md exact check: cards/g2_rust/g2_ghs_check.rs (standalone `rustc -O`, u128 arithmetic mod p) with its output beside it, 31 checks passing.

- **Field and curves checked first.** p = 2^64 − 2^32 + 1, p ≡ 1 mod 5, z^5 − 3 irreducible, the Frobenius acts as z^p = ω·z, b = 263z is a non-square, EcGFp5 has exactly one rational 2-torsion point, its 2-isogenous curve again has exactly one, EcMasFp5 has none, and neither j-invariant is in F_p.
- **Cover genus, computed.** For the x-coordinate Kummer cover descended from F_{p^5} to F_p, no product of the Frobenius conjugates of f is a square (all 31 subsets tested), so m = 5. Branch points: 12 for EcGFp5 (0, ∞ and ten conjugate roots), 16 for EcMasFp5, giving genus 2^{m−1}(t/2 − 2) + 1 = 65 and 97. I re-derived both genera by hand from the same formula. The toy cases reproduce the known weak class ((d, n) = (5, 4), genus 5). G2 also constructed covers of genus 33 by Möbius changes and derived a minimum of 25 for EcGFp5 (derived, not constructed, not source-verified).
- **Cost.** Index calculus on a genus-25 Jacobian is about 2^207, and about 2^180 under the most optimistic Enge–Gaudry estimate; genus 33 is about 2^247. All are far above rho.
- **Gorla–Massierer** (frozen text): their n = 5, genus-5 remark is about trace-zero varieties of curves defined over the subfield; neither curve is one, as the design note already said.
- **Oracle-assisted SDH.** Joux–Vitse's learning phase needs about p/2 = 2^63 queries (about 293 years at 10^9 queries per second). Granger's slides give an asymptotic q^{5/6} ≈ 2^53.3 variant with unmeasured constants. Deployments found: Schnorr signatures (Lighter, POD2, goldilocks-crypto), a Miden ElGamal prototype, and an EcMasFp5 mention in a pil2-proofman PR; none exposes [d]X for attacker-chosen X.
- **Not read:** Iijima–Momose–Chao (ePrint 2015/805, 2013/487) and Diem's odd-characteristic GHS paper; ePrint returned 403 to both WebFetch and curl. The cover verdict rests on the computation and on secondary sources (Chao's ECC 2007 slides, Tian's arXiv summary).

5. The decisive experiment (specified, not run, not authorized here)

From G1, with my additions. It measures the one quantity every verdict above depends on.

- **What.** One complete m = 5 solve (F4 plus parametrisation) of the F_p-rational (Z/2)^4 ⋊ S5 quotient of `KN-OPEN-9b4a2b`, at three toy primes spanning at least 12 bits, after validating the quotient construction at n = m = 3 (predicted D = 16 against 64). Report D and c in F_p operations, with RSS.
- **Frozen criteria (G1).** PROMOTE if D ≤ 2^17 and c ≤ 2^40; CLOSE if c ≥ 2^44 or D ≥ 2^19; otherwise record not_measured with the RSS at death. Expected outcome: CLOSE, tuned total ≈ 2^{132-2}134.
- **Envelope.** 512 GiB RAM, 32 threads, at most 72 h per cell, no address-space limit below 480 GiB. Do not attempt the S5 system at m = 5: G1's dense bound reaches 9.3 TiB at degree 76.
- **Repository rules that apply before any run.** Frozen protocol and task card (experiment-director, then executor-mechanical); a resume checkpoint (`scripts/checkpoint.sh init`); x86 runs go through `isolab-eks` on EKS Spot; any claim needs the validation gate. None of this was started.

6. Corrections found

where	correction
Pornin §2 (the "≥ 2^142")	omits the 5! factor that Joux–Vitse carry (JV line 365); and the large-prime count is not the cheapest rebalancing when systems are expensive, so the figure is not a floor
atlas `LEAD-ICP-gfp5-symmetrized-gaudry discriminates`	"still far above rho's 2^160 only if the band collapses" inverts the picture: in raw time every band is far *below* rho; the live comparison is with the 128-bit target, and memory
atlas `LEAD-ICP-gfp5-oa-sdhp-jv`	"the (n−1)-point variant … is expected to cost more than rho": it is within ±2.5 bits of rho in raw time (G1); it closes by memory
`research/gfp5_deployed_curves_20260920.md` cost bands	already re-anchored by the C1 review (K7); both those bands and the design note's use the large-prime count

crypto-autoresearcher `inputs/PORNIN-2022-274-ECGFP5/source_record.yaml` line 25	says the modulus is z^5 − 2; the paper chooses z^5 − 3 (z^5 ± 2 is reducible because 2 is a fifth-power residue mod p, checked)
my previous chat answer	quoted the atlas's "unmeasured 2^121" symmetrized prediction as a live figure; the C1 review (K7) had already shown that band mis-anchored by 5! and re-anchored it to ≥ 2^134.4 (model). The tuned-base figures above supersede both

7. What I verified and what I did not

Verified by me: the quotations from Pornin and Joux–Vitse (line numbers above); the benchmark constants (Pornin's table: F_p multiplication 10.18 cycles, point addition 1,328.50 cycles); my own cost model and the c^{1/3} scaling; agreement of my model with G1's structure; the repository records cited in §2.

Also verified by me: G2's Rust check passes all 31 checks; the two canonical cover genera (65, 97) recomputed by hand; the modulus z^5 − 3 against the paper text.

Relayed from evaluators: G1's word-operation constants, memory figures, F4 RAM extrapolation and crossing points; G2's constructed genus-33 covers, the genus-25 minimum, the Jacobian index-calculus costs, the oracle-assisted SDH figures and the deployment search.

Not verified: Gaudry 2009 full text (paywalled), the exponents of single and double large primes beyond Joux–Vitse's quotation, any 64-bit F4 timing (none exists; the 32-bit $\tilde{c}$ is carried over), the repository's §5a GAE band (absent from the frozen checkout), and anything about deployed protocol use beyond G2's search.

8. Files

`README.md`, `report.pdf`, `figures/cost_landscape.{png,svg}`, `cost_landscape.{py,json,md}`, `plot_cost_landscape.py`, `cards/` (shared brief, the two evaluator cards, G1's scripts, G2's Rust check), `build.sh`.